

September 2026 Exam

EC400

Probability and Statistical Inference

Sample Exam 1

Suitable for all candidates

Instructions to candidates

This paper contains **SIX** questions. Answer **ANY FOUR** questions.

Each question carries **25 marks**; the four questions attempted therefore carry **100 marks** in total.

Show your reasoning and working. State clearly any additional assumption that you use.

If more than four questions are attempted, the four answers awarded the highest marks will count toward the examination mark.

Time Allowed	Reading Time: 15 minutes
	Writing Time: 2 hours

You are supplied with:	<i>Standard Normal and Student's t tables</i>
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You may also use:	<i>No additional materials</i>
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Calculators:	<i>Calculators are NOT allowed in this examination</i>
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Question 1

Let $X_1, \dots, X_n$ be iid random variables with $\mathbb{E}[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2 < \infty$, where $n \geq 2$. Define

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i, \quad S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2.$$

[25 marks]

(a) Prove the identity

$$\sum_{i=1}^n (X_i - \bar{X}_n)^2 = \sum_{i=1}^n (X_i - \mu)^2 - n(\bar{X}_n - \mu)^2.$$

[5 marks]

(b) Use part (a) to show that S_n^2 is unbiased for σ^2 . Explain why the denominator is $n - 1$ rather than n . **[5 marks]**

(c) Define the variance estimator with divisor n by

$$V_n = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2.$$

Derive its bias. If $v_n = \text{Var}(S_n^2) < \infty$, express the MSEs of V_n and S_n^2 in terms of n , σ^2 and v_n . Explain the bias-variance trade-off. **[6 marks]**

(d) Assume additionally that $\mathbb{E}[X_i^4] < \infty$. Starting from the identity

$$S_n^2 = \frac{n}{n-1} \left[\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2 - (\bar{X}_n - \mu)^2 \right],$$

use the law of large numbers (LLN), the union bound and the probability bound below to prove that $S_n^2 \xrightarrow{p} \sigma^2$. Explain why an estimator can be unbiased without being consistent, or consistent without being unbiased. You may use, for any finite-variance random variable Z and any $t > 0$,

$$\Pr(|Z - \mathbb{E}[Z]| > t) \leq \frac{\text{Var}(Z)}{t^2}.$$

You may also use the following consequence of the displayed identity:

$$\begin{aligned} |S_n^2 - \sigma^2| &\leq \frac{n}{n-1} \left| \frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2 - \sigma^2 \right| \\ &\quad + \frac{n}{n-1} (\bar{X}_n - \mu)^2 + \frac{\sigma^2}{n-1}. \end{aligned}$$

[9 marks]

Question 2

Let D be binary, with $\Pr(D = 1) = p$, where $0 < p < 1$. Conditional on D , a continuous random variable X has density

$$f_{X|D}(x | 1) = 2x, \quad f_{X|D}(x | 0) = 2(1 - x), \quad 0 < x < 1,$$

and both conditional densities are zero otherwise.

[25 marks]

- (a) Verify that the two conditional densities are valid and derive the marginal density $f_X(x)$ over the whole real line. **[5 marks]**
- (b) Use Bayes' rule for densities to derive $\Pr(D = 1 | X = x)$ for $0 < x < 1$. Prove that this posterior probability is strictly increasing in x , and interpret the result. **[5 marks]**
- (c) For an arbitrary $c \in (0, 1)$, derive $\Pr(D = 1 | X > c)$ and prove that it exceeds the prior probability p . **[4 marks]**
- (d) Calculate the conditional means and variances of X given each value of D . Then use the laws of iterated expectations and total variance to find $\mathbb{E}[X]$ and $\text{Var}(X)$. **[6 marks]**
- (e) Calculate $\text{Cov}(D, X)$. Are D and X independent? Justify your conclusion. **[5 marks]**

Question 3

Let X_1, X_2, X_3 be iid $N(\mu, \sigma^2)$ random variables, where $\sigma^2 > 0$, and define

$$A = \frac{X_1 + X_2 + X_3}{3}, \quad B = \frac{X_1 - X_2}{\sqrt{2}}, \quad C = \frac{X_1 + X_2 - 2X_3}{\sqrt{6}}.$$

You may use these facts: a jointly Normal vector with a diagonal covariance matrix has independent entries, and functions of independent groups of variables remain independent. **[25 marks]**

- (a) Find the marginal distributions of A, B and C . **[5 marks]**
- (b) Calculate $\text{Cov}(A, B)$, $\text{Cov}(A, C)$ and $\text{Cov}(B, C)$. Deduce that A, B, C are independent, stating why these three zero covariances are sufficient in this case. **[6 marks]**
- (c) Find the distribution of

$$Q = \frac{B^2 + C^2}{\sigma^2}.$$

[4 marks]

- (d) Find the distribution of

$$T = \frac{\sqrt{3}(A - \mu)/\sigma}{\sqrt{Q/2}}.$$

State every distributional and independence fact that you use.

[6 marks]

- (e) Define

$$D = \frac{B + C}{\sqrt{2}}, \quad E = \frac{B - C}{\sqrt{2}}.$$

Find their marginal distributions and covariance, and deduce that they are independent. Hence find the distributions of D/E and $(D/E)^2$. **[4 marks]**

Question 4

Four pupils are observed in two independently sampled matched pairs, (1, 2) and (3, 4). Within each pair one pupil receives a tutoring programme. Let $z_i = -1$ for control and $z_i = 1$ for treatment, with

$$z = (-1, 1, -1, 1)', \quad y_i = \alpha + \beta z_i + \varepsilon_i, \quad \mathbb{E}[\varepsilon | X] = 0.$$

Conditional on the treatment design,

$$\Omega = \text{Var}(\varepsilon | X) = \begin{pmatrix} s_1^2 & \rho_1 s_1^2 & 0 & 0 \\ \rho_1 s_1^2 & s_1^2 & 0 & 0 \\ 0 & 0 & s_2^2 & \rho_2 s_2^2 \\ 0 & 0 & \rho_2 s_2^2 & s_2^2 \end{pmatrix}.$$

Here s_1^2, s_2^2 are the error variances in the two pairs, and ρ_1, ρ_2 are the correlations within those pairs. Assume $s_1^2, s_2^2 > 0$ and $-1 < \rho_1, \rho_2 < 1$. **[25 marks]**

- (a) Let $m_0 = \mathbb{E}[y_i | z_i = -1]$ and $m_1 = \mathbb{E}[y_i | z_i = 1]$. Express m_0, m_1, α and β in terms of one another. **[4 marks]**
- (b) Suppose treatment is randomly assigned within each pair, with each pupil having probability $1/2$ of treatment. Let $D = (z + 1)/2$ and assume $Y = DY(1) + (1 - D)Y(0)$, where $Y(1), Y(0)$ are the potential outcomes. What causal quantity does 2β identify in the experimental population? Does random assignment alone make that population representative of all pupils? **[4 marks]**
- (c) Let

$$\bar{y}_0 = \frac{y_1 + y_3}{2}, \quad \bar{y}_1 = \frac{y_2 + y_4}{2}.$$

Using the fact that least-squares fitted values equal the two group sample means in this regression with an intercept and a binary regressor, derive $\hat{\alpha}$ and $\hat{\beta}$. **[5 marks]**

- (d) Describe the heteroscedasticity and dependence represented by Ω . In a large study with many independent matched pairs of this form, which standard-error method would naturally accommodate both features? **[5 marks]**
- (e) Calculate $\text{Var}(\bar{y}_0 | X)$, $\text{Var}(\bar{y}_1 | X)$ and $\text{Cov}(\bar{y}_0, \bar{y}_1 | X)$. Hence derive the conditional variance of $\hat{\beta}$. Explain how the positive within-pair covariances affect the variance of $\hat{\beta}$ relative to a calculation that incorrectly sets the off-diagonal elements of Ω to zero. **[7 marks]**

Question 5

An education researcher observes m independently sampled classes, with $r \geq 2$ pupils observed in each class. For class $j = 1, \dots, m$ and pupil $k = 1, \dots, r$, suppose

$$Y_{jk} = \mu + A_j + e_{jk},$$

where $A_j \sim N(0, \tau^2)$ are iid class shocks, $e_{jk} \sim N(0, \sigma^2)$ are iid pupil shocks, all class and pupil shocks are independent, and $\tau^2, \sigma^2 > 0$. **[25 marks]**

- (a) Find the mean and variance of Y_{jk} . Find the covariance between two distinct pupils in the same class and between pupils in different classes. Hence calculate the correlation between two pupils in the same class. Recall that correlation is covariance divided by the product of the two standard deviations. **[5 marks]**

(b) Let

$$\bar{Y} = \frac{1}{mr} \sum_{j=1}^m \sum_{k=1}^r Y_{jk}$$

be the overall sample mean. Express $\bar{Y} - \mu$ in terms of the class and pupil shocks, and derive its exact variance. **[6 marks]**

(c) Use mean-square convergence to show that $\bar{Y}$ is consistent for μ as $m \rightarrow \infty$. **[4 marks]**

(d) Compare the correct standard error of $\bar{Y}$ with the standard error obtained by incorrectly treating the mr pupil outcomes as iid. Express their ratio in terms of r and the correlation ρ between pupils in the same class. Prove that the iid calculation understates uncertainty and explain the source of the error. **[6 marks]**

(e) Compare the limits obtained by (i) letting $r \rightarrow \infty$ with m fixed and (ii) letting $m \rightarrow \infty$ with r fixed. In which sequence is the overall sample mean consistent for μ ? Explain the distinction in terms of the independent units in the study. **[4 marks]**

Question 6

An estimator of $\beta = (\beta_1, \beta_2)'$ is approximately Normal with estimate and estimated covariance matrix

$$\hat{\beta} = \begin{pmatrix} 0.18 \\ -0.36 \end{pmatrix}, \quad \hat{V} = \begin{pmatrix} 0.01 & 0 \\ 0 & 0.04 \end{pmatrix}.$$

For the calculations below, you may use $\Phi(1.80) = 0.9641$, the two-sided 5% standard-Normal critical value 1.96, the 5% critical value 5.991 for χ_2^2 , and $\Pr(\chi_2^2 \geq 6.48) = 0.0392$. You may also use $\sqrt{0.05} \simeq 0.224$. **[25 marks]**

(a) For the two individual null hypotheses $H_{01} : \beta_1 = 0$ and $H_{02} : \beta_2 = 0$, calculate the z statistics and two-sided p -values. Test each null separately at the unadjusted 5% level. **[5 marks]**

(b) Construct the two separate 95% confidence intervals and state whether each contains zero. **[5 marks]**

(c) We now want to test whether *both* coefficients are zero: $H_0 : \beta = \beta_0 = (0, 0)'$. This is a *joint* null hypothesis. A *Wald test* extends the distance-from-the-null idea in Lecture 4 to several coefficients, allowing for their variances and covariances. For this question, use the statistic

$$W = (\hat{\beta} - \beta_0)' \hat{V}^{-1} (\hat{\beta} - \beta_0).$$

You may use the fact that, under this joint null, W is approximately χ_2^2 : there are two restrictions being tested. Calculate W and test at the 5% level, using both the supplied critical value and tail probability. **[6 marks]**

(d) Compare the joint decision in part (c) with the separate decisions in part (a). Explain why they can differ. Does rejecting the joint null tell us that each coefficient is nonzero? Does failing to reject an individual null prove that the corresponding coefficient is zero? **[4 marks]**

(e) Using a standard-Normal reference distribution, test each of the following null hypotheses:

$$H_{0a} : 2\beta_1 + \beta_2 = 0, \quad H_{0b} : \beta_1 - \beta_2 = 0.$$

Explain why these decisions need not agree with part (c): what does each null hypothesis ask? **[5 marks]**

END OF PAPER

TABLE FD.1 Observations on Income and Education, 20 Observations

I = Observation,
 Y = Income.

Source: Data are artificial.

APPENDIX G



STATISTICAL TABLES

TABLE G.1 Cumulative Normal Distribution. Table Entry Is $\Phi(z) = \text{Prob}[Z \leq z]$

[illegible]

TABLE G.2 Percentiles of the Student's t Distribution. Table Entry Is x Such that $\text{Prob}[t_n \leq x] = P$

n	.750	.900	.950	.975	.990	.995
1	1.000	3.078	6.314	12.706	31.821	63.657
2	.816	1.886	2.920	4.303	6.965	9.925
3	.765	1.638	2.353	3.182	4.541	5.841
4	.741	1.533	2.132	2.776	3.747	4.604
5	.727	1.476	2.015	2.571	3.365	4.032
6	.718	1.440	1.943	2.447	3.143	3.707
7	.711	1.415	1.895	2.365	2.998	3.499
8	.706	1.397	1.860	2.306	2.896	3.355
9	.703	1.383	1.833	2.262	2.821	3.250
10	.700	1.372	1.812	2.228	2.764	3.169
11	.697	1.363	1.796	2.201	2.718	3.106
12	.695	1.356	1.782	2.179	2.681	3.055
13	.694	1.350	1.771	2.160	2.650	3.012
14	.692	1.345	1.761	2.145	2.624	2.977
15	.691	1.341	1.753	2.131	2.602	2.947
16	.690	1.337	1.746	2.120	2.583	2.921
17	.689	1.333	1.740	2.110	2.567	2.898
18	.688	1.330	1.734	2.101	2.552	2.878
19	.688	1.328	1.729	2.093	2.539	2.861
20	.687	1.325	1.725	2.086	2.528	2.845
21	.686	1.323	1.721	2.080	2.518	2.831
22	.686	1.321	1.717	2.074	2.508	2.819
23	.685	1.319	1.714	2.069	2.500	2.807
24	.685	1.318	1.711	2.064	2.492	2.797
25	.684	1.316	1.708	2.060	2.485	2.787
26	.684	1.315	1.706	2.056	2.479	2.779
27	.684	1.314	1.703	2.052	2.473	2.771
28	.683	1.313	1.701	2.048	2.467	2.763
29	.683	1.311	1.699	2.045	2.462	2.756
30	.683	1.310	1.697	2.042	2.457	2.750
35	.682	1.306	1.690	2.030	2.438	2.724
40	.681	1.303	1.684	2.021	2.423	2.704
45	.680	1.301	1.679	2.014	2.412	2.690
50	.679	1.299	1.676	2.009	2.403	2.678
60	.679	1.296	1.671	2.000	2.390	2.660
70	.678	1.294	1.667	1.994	2.381	2.648
80	.678	1.292	1.664	1.990	2.374	2.639
90	.677	1.291	1.662	1.987	2.368	2.632
100	.677	1.290	1.660	1.984	2.364	2.626
∞	.674	1.282	1.645	1.960	2.326	2.576